Logos Reference Manual

A Logic for Interpreted and Verified AI Reasoning

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Primary References:

"The Construction of Possible Worlds", Brast-McKie, (under review), 2025.

"Counterfactual Worlds", Brast-McKie, *J. Phil. Logic*, 2025.

"Identity and Aboutness", Brast-McKie, *J. Phil. Logic*, 2021.

Abstract

This reference manual provides the formal specification of the Logos which is an extensible formal language of though designed to train AI systems in verified logical reasoning. The Logos is interpreted by a hyperintensional task semantics designed to support operators for reasoning about necessity, possibility, time, and counterfactual conditionals. This semantics and its corresponding proof theory is successively extended by a plugin system to support a wide range of additional operators for complex reasoning to plan and evaluate actions in multi-agent systems under conditions of uncertainty.

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1 Introduction

This reference manual provides the formal specification for the Logos proof system, semantics, metalogic, and library of theorems. The semantics is organized around the semantic frames that provides the primitive structure used to interpret the logical operators included in the Logos. By successively extending the primitive resources included in a frame, it becomes possible to provide semantic clauses for a new range of operators, thereby extending the expressive power of the Logos for systematic reasoning.

The Logos is interpreted over a hyperintensional *state semantics* which comprises the *Constitutive Foundation*. By contrast to intensional semantic theories which take worlds to be primitive, the state semantics takes states and their parthood relations to be primitive to provide fine-grained semantic contents by which to recursively interpret the primitive expressions of the language. The *Dynamical Foundation* extends the state semantics to include a *task relation* parameterized by durations belonging to a *temporal order* in order to govern the lawful transitions by which a system and its parts evolve. These ingredients are then used to construct *world sates* which represent total instantaneous states of the system under study, as well as *world histories* which model the possible evolutions of that system by mapping durations to world states. For instance, in modeling a game of chess, the world states correspond to the legal complete configurations of the chessboard, and the world histories correspond to the possible games of chess.

Given a world history, each duration may be referred to as a *time* in that history. Together, a world history and time suffice to recursively assign truth-values to all sentences of the foundational layers of the Logos. In addition to the standard Boolean and first-order operators, the foundational semantic layers for the Logos support modal, temporal, counterfactual, constitutive, and causal operators for complex reasoning including planning and evaluating past and future actions. Modular plugins are then provided to extend the expressive power of the foundational layers of the Logos to include sets of related logical operators. In addition to extending the semantic primitives included in a frame, plugins may contribute new contextual parameters at which sentences are evaluated. For instance, given the addition of a *credence function* which assigns likelihoods to propositions, a *belief parameter* is required along with a world history and time to provide a semantics for the belief operator. Given a world history, time, and belief parameter, a proposition is believed just in case the credence in that proposition is greater than the belief parameter included in the point of evaluation.

1.1 Extension Dependencies

This section presents the layer extensions by which the Logos is constructed where Figure 1 shows the dependency structure among extensions where the first two layers provide an essential foundation followed by a modular plugin system for language extensions.

Whereas the Constitutive Foundation and Explanatory Extension form the required base, the middle extensions (Epistemic, Abilities, Normative, Choice, Spatial, and others) are modular plugins that can be combined in any subset. The Agential Extension, Social Extension, and Reflection Extension all inherit from the layers above and are free to interact with each other. The Agential Extension provides third-person multi-agent reasoning, the Social Extension provides the construction of the common ground and trust relations, and the Reflection Extension enables first-person metacognition.

1.2 Layer Descriptions

1. **Constitutive Foundation**: Provides the hyperintensional base with predicates (F, G), functions (f, g), variables (x, y), booleans ($\neg$, $\wedge$, $\vee$), extremal ($\bot$, $\top$), lambda abstraction (λ), identity ($=$), and constitutive operators (propositional identity $\equiv$, grounding $\leqslant$, essence $\sqsubseteq$, reduction $\Rightarrow$) over a complete lattice with bilateral propositions (verifier/falsifier pairs). This enables fine-grained distinctions between propositions with identical truth-values and modal profiles but different verification conditions, essential for exact specification of what makes a claim true.

2. **Dynamical Foundation**: Adds quantifiers ($\forall$, $\exists$), modals ($\Box$, $\Diamond$) for necessity and possibility, tense (H, G, P, F, $\lhd$, $\rhd$) for reasoning across time, store/recall operators ($\downarrow$, $\uparrow$) for temporal reference, actuality (Act) for distinguishing actual from possible, counterfactual conditionals ($\Box\!\!\rightarrow$, $\Diamond\!\!\rightarrow$) for hypothetical reasoning, and causal operators ($\bigcirc\!\!\rightarrow$, $\odot\!\!\rightarrow$) for explanatory links. Together these resources enable reasoning about action sequences, future contingency, what would have happened under different circumstances, and causal relationships.

3. **Epistemic Extension**: Introduces epistemic modals (Mi, Mu), belief (B), probability (Pr), and indicative conditional operators ($\hookrightarrow$) for reasoning under uncertainty. By extending the semantic framework with credence functions to assign weights to state transitions, agents can track how evidence updates beliefs across time and distinguish what an agent believes from what is actually the case.

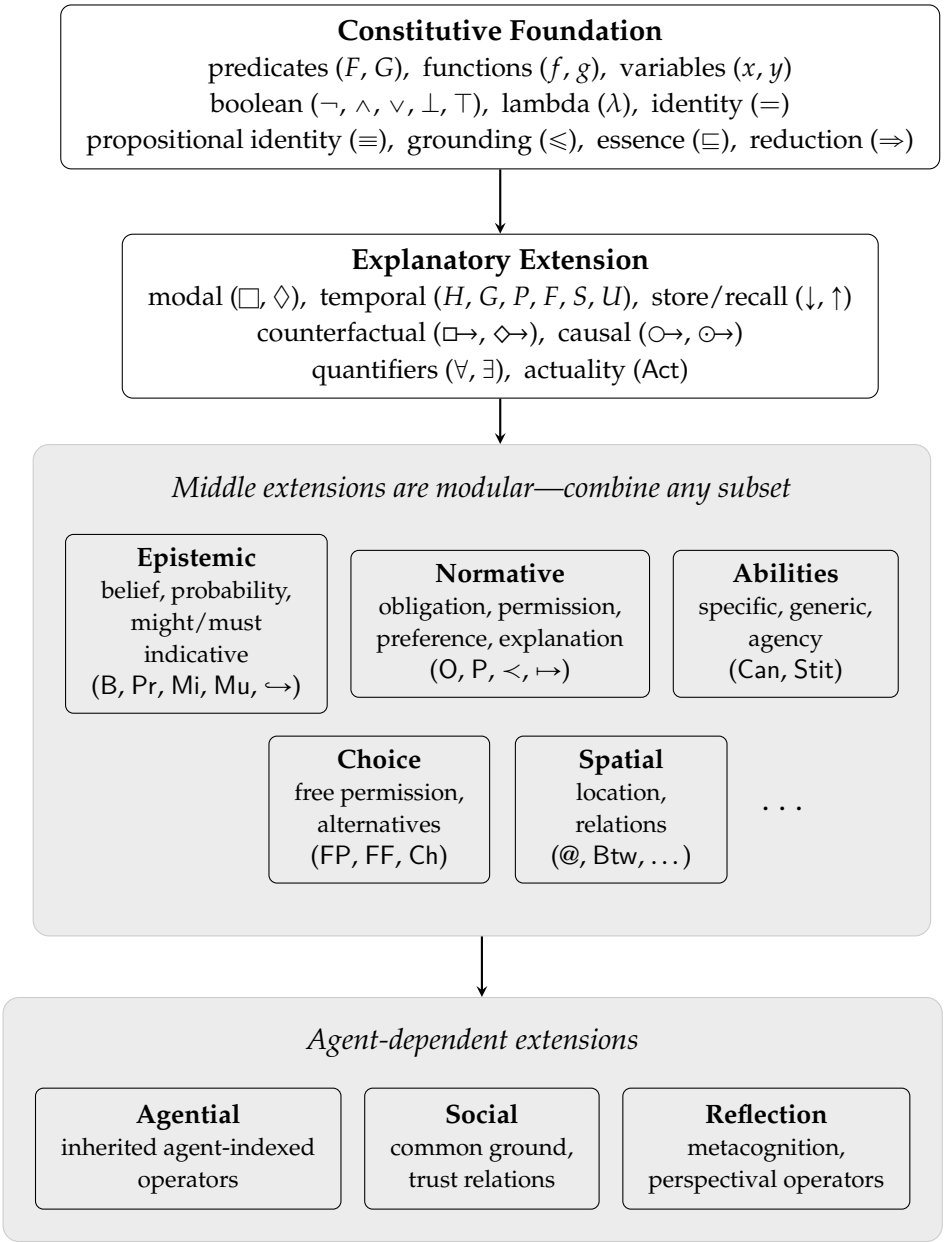

Figure 1: Extension dependency structure in the Logos system.

4. **Abilities Extension**: Provides ability (Can) and STIT (stit) operators for reasoning about capacities. Ability modals express what an agent can bring about through their capacities, with stronger truth conditions than circumstantial possibility since ability entails possibility but not *vice versa*. This enables reasoning about what agents are capable of accomplishing given their intrinsic capacities and circumstances.

5. **Normative Extension**: Adds obligation (O), permission (P), and a preference operator ($\prec$) with value orderings over states for reasoning about what ought to be done, what is permitted, and how to rank alternatives. This enables ethical reasoning about alternatives.

6. **Choice Extension**: Introduces free choice permission (FP) and best choice (Ch) operators for reasoning about permission over alternatives, enabling reasoning about which alternatives are permitted and how permissions combine.

7. **Spatial Extension**: Provides location-dependent operators and spatial relations for reasoning about where things are and they are oriented relative to an orientation basis. This supports navigation, containment reasoning, and location-sensitive planning.

8. **Agential Extension**: Indexes epistemic and normative operators to specific agents (B_a, $\prec$, O_a, ...), enabling reasoning about what different agents believe, prefer, and are obligated to do. This transforms single-perspective reasoning into multi-agent coordination where agents can model each other's epistemic and normative states.

9. **Social Extension**: Provides resources for common ground construction and group-level reasoning, building upon the Agential Extension to enable reasoning about shared beliefs, build trust in other agents, and coordinate on common goals.

10. **Reflection Extension**: Adds first-person metacognitive operators (I) for self-directed reasoning about one's own beliefs, abilities, and progress towards goals. This enables an agent to reason about what it is in a good position to believe given evaluation of its own limitations and track relative to other agents. Essential for AI systems that must understand their own capabilities and benefit from the capacity to learn by the testimony of others.

1.3 Document Organization

This reference manual is organized as follows:

Section 2 presents the Constitutive Foundation, including the mereological state space, verification and falsification clauses, and bilateral propositions.

Section 3 introduces the syntactic primitives of the Explanatory Extension, including modal, temporal, and counterfactual operators.

Section 4 provides the semantic framework for the Explanatory Extension, including core frames, world-histories, and truth conditions.

Section 5 presents the axiom system for counterfactual logic.

...

Section 10 specifies the Reflection Extension for first-person metacognitive reasoning, including truth conditions, derived operators, and temporal interactions.

1.4 Lean Implementation

The Logos system is implemented in Lean 4 with Mathlib. Each definition in this reference has a corresponding Lean implementation, referenced using the `\leansrc{}{}` command.

For example, `Logos.Foundation.Frame.ConstitutiveFrame` refers to the `ConstitutiveFrame` definition in the `Logos.Foundation.Frame` module. The implementation can be found in the `ProofChecker` repository in the `Theories/Logos/` directory.

2 Constitutive Foundation

The Constitutive Foundation provides the foundational semantic structure based on exact truthmaker semantics. Evaluation is hyperintensional, distinguishing propositions that agree on truth-value across all possible worlds but differ in their exact verification and falsification conditions.

2.1 Syntactic Primitives

The Constitutive Foundation interprets the following syntactic primitives:

- **Variables**: $x, y, z, \ldots$ (ranging over states)
- **Individual constants**: $a, b, c, \ldots$ (0-place function symbols)
- **n-place function symbols**: $f, g, h, \ldots$
- **n-place predicates**: $F, G, H, \ldots$
- **Sentence letters**: $p, q, r, \ldots$ (0-place predicates)
- **Lambda abstraction**: $\lambda x.\varphi$ (binding variable x in formula φ)
- **First-order identity**: $=$ (identity between terms)
- **Logical connectives**: $\neg, \wedge, \vee, \top, \bot, \equiv$

2.2 Constitutive Frame

Definition 2.1. A *constitutive frame* is a structure $\mathbf{F} = \langle S, \sqsubseteq \rangle$ where:

- S is a nonempty set of *states*
- $\sqsubseteq$ is a partial order on S where $\langle S, \sqsubseteq \rangle$ is a complete lattice

Remark 2.2. The constitutive frame is non-modal: neither possibility nor compatibility can be defined since this requires the task relation introduced in the Explanatory Extension, and taking S itself to consist of *possible* states disrupts closure under fusion. This makes the conjunction of incompatible propositions empty, stripping the semantics of its capacity to distinguish $p \wedge \neg p$ and $p \wedge \neg q$, undermining the ability to distinguish propositions differing in subject-matter with no gain in elegance or simplicity.

Remark 2.3. The lattice structure provides:

- **Null state** $\square$: The bottom element (fusion of the empty set)
- **Full state** $\blacksquare$: The top element (fusion of all states)
- **Fusion** $s \sqcup t$: The least upper bound of s and t

See `Logos.Foundation.Frame.ConstitutiveFrame` for the implementation.

2.3 Constitutive Model

Definition 2.4. A *constitutive model* is a structure $\mathcal{M} = \langle S, \sqsubseteq, |\cdot| \rangle$ where $\langle S, \sqsubseteq \rangle$ is a constitutive frame and $|\cdot|$ is an *interpretation function* satisfying:

- **n-place function symbols** $|f| : (\texttt{Fin}\ n \to S) \to S$
- **0-place function symbols** $|c| : (\texttt{Fin}\ 0 \to S) \to S$
- **n-place predicates** $|F| = \langle |F|^+, |F|^- \rangle$ where:
  - $|F|^+ = (\texttt{Fin}\ n \to S) \to S$ is the *verifier function type*
  - $|F|^- = (\texttt{Fin}\ n \to S) \to S$ is the *falsifier function type*
- **Sentence letters** $|p| = \langle |p|^+, |p|^- \rangle$ of verifier and falsifier state sets

See `Logos.Foundation.Model.ConstitutiveModel` for the implementation.

2.4 Variable Assignment

Definition 2.5. A *variable assignment* σ is a function $\sigma : \mathrm{Var} \to S$

Notation 2.6. Greek letters ($\tau, \alpha, \beta, \ldots$) are reserved for world-histories. The letter σ (with subscripts $\sigma_1, \sigma_2, \ldots$) denotes variable assignments, chosen for compatibility across LaTeX, markdown, and Lean notation.

Definition 2.7. The *x-variant* $\sigma[v/x]$ of the variable assignment σ differs at most from σ by assigning x to v.

Definition 2.8. The *extension* of a term given a model $\mathcal{M}$ and assignment σ:

- **Variable** $[\![x]\!]^\sigma_{\mathcal{M}} = \sigma(x)$
- **Function application** $[\![f(t_1, \ldots, t_n)]\!]^\sigma_{\mathcal{M}} = |f|(\lambda i : \texttt{Fin}\ n, [\![t_i]\!]^\sigma_{\mathcal{M}})$

See `Logos.Foundation.Assignment.Assignment` for the implementation.

2.5 Verification and Falsification

The *verification* ($\Vdash$) and *falsification* ($\dashv\!\Vert$) of a formula φ is defined recursively relative to a model $\mathcal{M}$, assignment σ, and state s.

2.5.1 Atomic Formulas

Definition 2.9 (Atomic Verification and Falsification).

$$\mathcal{M}, \sigma, s \Vdash F(t_1, \ldots, t_n) \ \textit{iff} \ \exists i : \texttt{Fin}\ (\#^+(F)) : s = \mathrm{verfn}_F(i)(\lambda j : \texttt{Fin}\ n, [\![t_j]\!]^\sigma_{\mathcal{M}}) \tag{1}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vert F(t_1, \ldots, t_n) \ \textit{iff} \ \exists i : \texttt{Fin}\ (\#^-(F)) : s = \mathrm{faln}_F(i)(\lambda j : \texttt{Fin}\ n, [\![t_j]\!]^\sigma_{\mathcal{M}}) \tag{2}$$

2.5.2 Lambda Abstraction

Definition 2.10 (Lambda Verification and Falsification).

$$\mathcal{M}, \sigma, s \Vdash (\lambda x.\varphi)(t) \ \textit{iff} \ \mathcal{M}, \sigma[[\![t]\!]^\sigma_{\mathcal{M}}/x], s \Vdash \varphi \tag{3}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vert (\lambda x.\varphi)(t) \ \textit{iff} \ \mathcal{M}, \sigma[[\![t]\!]^\sigma_{\mathcal{M}}/x], s \dashv\!\Vert \varphi \tag{4}$$

2.5.3 Negation

Definition 2.11 (Negation Verification and Falsification).

$$\mathcal{M}, \sigma, s \Vdash \neg\varphi \ \textit{iff} \ \mathcal{M}, \sigma, s \dashv\!\Vert \varphi \tag{5}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vert \neg\varphi \ \textit{iff} \ \mathcal{M}, \sigma, s \Vdash \varphi \tag{6}$$

2.5.4 Conjunction

Definition 2.12 (Conjunction Verification and Falsification).

$$\mathcal{M}, \sigma, s \Vdash \varphi \wedge \psi \ \textit{iff} \ s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \Vdash \varphi \text{ and } \mathcal{M}, \sigma, u \Vdash \psi \tag{7}$$

$$\begin{aligned} \mathcal{M}, \sigma, s \dashv\!\Vert \varphi \wedge \psi \ \textit{iff} \ & \mathcal{M}, \sigma, s \dashv\!\Vert \varphi, \text{ or } \mathcal{M}, \sigma, s \dashv\!\Vert \psi, \text{ or} \\ & s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \dashv\!\Vert \varphi \text{ and } \mathcal{M}, \sigma, u \dashv\!\Vert \psi \end{aligned} \tag{8}$$

2.5.5 Disjunction

Definition 2.13 (Disjunction Verification and Falsification).

$$\begin{aligned} \mathcal{M}, \sigma, s \Vdash \varphi \vee \psi \ \textit{iff} \ & \mathcal{M}, \sigma, s \Vdash \varphi, \text{ or } \mathcal{M}, \sigma, s \Vdash \psi, \text{ or} \\ & s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \Vdash \varphi \text{ and } \mathcal{M}, \sigma, u \Vdash \psi \end{aligned} \tag{9}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vert \varphi \vee \psi \ \textit{iff} \ s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \dashv\!\Vert \varphi \text{ and } \mathcal{M}, \sigma, u \dashv\!\Vert \psi \tag{10}$$

2.5.6 Top and Bottom

Definition 2.14 (Top and Bottom Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash \top \text{ for all } s \in S \tag{11}$$
$$\mathcal{M}, \sigma, s \dashv\!\Vdash \top \ \textit{iff}\ s = \blacksquare \text{ (full state)} \tag{12}$$
$$\mathcal{M}, \sigma, s \Vdash\!\!\!\vdash \bot \text{ for all } s \tag{13}$$
$$\mathcal{M}, \sigma, s \dashv\!\Vdash \bot \ \textit{iff}\ s = \square \text{ (null state)} \tag{14}$$

2.5.7 Propositional Identity

Definition 2.15 (Propositional Identity Verification and Falsification)**.**

$$\begin{aligned}\mathcal{M}, \sigma, s \Vdash \varphi \equiv \psi \ \textit{iff}\ s = \square \text{ and } &\{t : \mathcal{M}, \sigma, t \Vdash \varphi\} = \{t : \mathcal{M}, \sigma, t \Vdash \psi\} \\ \text{and } &\{t : \mathcal{M}, \sigma, t \dashv\!\Vdash \varphi\} = \{t : \mathcal{M}, \sigma, t \dashv\!\Vdash \psi\}\end{aligned} \tag{15}$$
$$\begin{aligned}\mathcal{M}, \sigma, s \dashv\!\Vdash \varphi \equiv \psi \ \textit{iff}\ s = \square \text{ and } &(\{t : \mathcal{M}, \sigma, t \Vdash \varphi\} \neq \{t : \mathcal{M}, \sigma, t \Vdash \psi\} \\ \text{or } &\{t : \mathcal{M}, \sigma, t \dashv\!\Vdash \varphi\} \neq \{t : \mathcal{M}, \sigma, t \dashv\!\Vdash \psi\})\end{aligned} \tag{16}$$

See `Logos.Foundation.Semantics.verifies` for the Lean implementation.

2.6 Essence and Ground

These constitutive relations can be defined via propositional identity:

Definition 2.16 (Essence and Ground)**.**

$$\varphi \sqsubseteq \psi := \varphi \wedge \psi \equiv \psi \qquad \text{“}\varphi \text{ is essential to } \psi\text{” (conjunctive part)} \tag{17}$$
$$\varphi \leqslant \psi := \varphi \vee \psi \equiv \psi \qquad \text{“}\varphi \text{ grounds } \psi\text{” (disjunctive part)} \tag{18}$$

Remark 2.17 (Negation Distribution). These relations are interrelated through negation:

- $\varphi \sqsubseteq \psi$ iff $\neg\varphi \leqslant \neg\psi$
- $\varphi \leqslant \psi$ iff $\neg\varphi \sqsubseteq \neg\psi$

Remark 2.18 (Bilattice Structure). The space of bilateral propositions forms a non-interlaced bilattice where:

- $\langle P, \sqsubseteq \rangle$ and $\langle P, \leqslant \rangle$ are complete lattices
- Negation exchanges the two orderings: $X \leqslant Y$ iff $\neg X \sqsubseteq \neg Y$

- Conjunction and disjunction are the least upper bounds with respect to $\sqsubseteq$ and $\leqslant$ respectively

See `Logos.Foundation.Relations.essence` for the Lean implementation.

2.7 Bilateral Propositions

Definition 2.19 (Bilateral Proposition)**.** A *bilateral proposition* is an ordered pair $\langle V, F\rangle$ where:

- V and F are subsets of S closed under fusion
- $\langle V, F\rangle$ is *exclusive*: states in V are incompatible with states in F
- $\langle V, F\rangle$ is *exhaustive*: every possible state is compatible with some state in V or F

Definition 2.20 (Propositional Operations)**.**

$$X \otimes Y := \{s \sqcup t : s \in X, t \in Y\} \quad \text{(Product)} \tag{19}$$
$$X \oplus Y := X \cup Y \cup (X \otimes Y) \quad \text{(Sum)} \tag{20}$$
$$\langle V, F\rangle \wedge \langle V', F'\rangle := \langle V \otimes V', F \oplus F'\rangle \quad \text{(Conjunction)} \tag{21}$$
$$\langle V, F\rangle \vee \langle V', F'\rangle := \langle V \oplus V', F \otimes F'\rangle \quad \text{(Disjunction)} \tag{22}$$
$$\neg\langle V, F\rangle := \langle F, V\rangle \quad \text{(Negation)} \tag{23}$$

See `Logos.Foundation.Proposition.BilateralProp` for the Lean implementation.

2.8 Logical Consequence

Definition 2.21 (Logical Consequence (Constitutive Foundation))**.** Logical consequence at the Constitutive Foundation is restricted to propositional identity sentences:

$$\Gamma \vDash \varphi \textit{ iff } \text{for any model } \mathcal{M} \text{ and assignment } \sigma : \text{if } \mathcal{M}, \sigma, \square \Vdash \psi \text{ for all } \psi \in \Gamma, \text{ then } \mathcal{M}, \sigma, \square \Vdash \varphi$$

That is, φ is a consequence of Γ iff the null state verifies φ in any model where it verifies all premises.

Remark 2.22 (Identity Sentences and Evaluation Overlap). Identity sentences are formed from extensional (non-identity) sentences: $\varphi \equiv \psi$ where φ and ψ are atomic sentences or built from $\neg, \wedge, \vee$. The logical consequences holding

between identity sentences are preserved by further extensions. However, the Constitutive Foundation lacks the semantic resources to evaluate non-identity sentences, which depend on contingent states of affairs rather than purely structural relations in state space. The Constitutive Foundation is nevertheless important for defining a logic of propositional identity. The same theorems valid at this level remain valid in the Explanatory Extension, though the Explanatory Extension's definition of logical consequence differs, quantifying over world-histories and times in addition to models and variable assignments.

3 Explanatory Extension: Syntactic Primitives

The Explanatory Extension extends the Constitutive Foundation with temporal structure and a task relation, enabling evaluation of truth relative to world-histories and times. While the hyperintensional foundation remains (distinguishing propositions by their exact verifiers and falsifiers), this extension adds intensional evaluation relative to contextual parameters (world-history, time, variable assignment) to determine truth-values for all Explanatory Extension sentences.

3.1 Additional Syntactic Primitives

The Explanatory Extension interprets the following additional syntactic primitives beyond those of the Constitutive Foundation:

- **Modal operators**: $\Box$ (necessity), $\Diamond$ (possibility)
- **Temporal operators**: H (always past), G (always future), P (some past), F (some future)
- **Extended temporal operators**: $\lhd$ (since), $\rhd$ (until)
- **Counterfactual conditional**: $\Box\!\!\rightarrow$ (would-counterfactual)
- **Store/recall operators**: $\uparrow^i$ (store), $\downarrow^i$ (recall)

3.2 Well-Formed Sentences

A well-formed sentence at this extension includes:

1. All Constitutive Foundation sentences (atomic formulas, $\neg$, $\wedge$, $\vee$, $\top$, $\bot$, $\equiv$)

2. Modal sentences: $\Box\varphi$, $\Diamond\varphi$
3. Temporal sentences: $H\varphi$, $G\varphi$, $P\varphi$, $F\varphi$
4. Extended temporal sentences: $\varphi \lhd \psi$, $\varphi \rhd \psi$
5. Counterfactual sentences: $\varphi \mathrel{\Box\!\!\rightarrow} \chi$
6. Indexed sentences: $\uparrow^i \varphi$, $\downarrow^i \varphi$
7. Universal quantification: $\forall x.\varphi$
8. Actuality predicate: $\mathsf{Act}(t)$

3.3 Operator Summary

Category	**Symbol**	**Reading**
Modal	$\Box\varphi$	"Necessarily φ"
	$\Diamond\varphi$	"Possibly φ"
Past Temporal	$H\varphi$	"It has always been that φ"
	$P\varphi$	"It was the case that φ"
Future Temporal	$G\varphi$	"It will always be that φ"
	$F\varphi$	"It will be the case that φ"
Extended	$\varphi \lhd \psi$	"φ since ψ"
	$\varphi \rhd \psi$	"φ until ψ"
Counterfactual	$\varphi \mathrel{\Box\!\!\rightarrow} \chi$	"If φ were the case, then χ"
Store/Recall	$\uparrow^i \varphi$	"Store current time and evaluate φ"
	$\downarrow^i \varphi$	"Recall stored time and evaluate φ"

Table 1: Explanatory Extension operators

3.4 Derived Operators

Definition 3.1 (Derived Temporal Operators)**.**

$$\triangle\varphi := H\varphi \wedge \varphi \wedge G\varphi \qquad \text{"Always } \varphi\text{" (at all times)} \tag{24}$$

$$\triangledown\varphi := P\varphi \vee \varphi \vee F\varphi \qquad \text{"Sometimes } \varphi\text{" (at some time)} \tag{25}$$

Definition 3.2 (Material Conditional)**.**

$$\varphi \to \psi := \neg\varphi \vee \psi$$

Definition 3.3 (Biconditional)**.**

$$\varphi \leftrightarrow \psi := (\varphi \to \psi) \wedge (\psi \to \varphi)$$

See `Logos.Explanatory.Syntax.CoreFormula` for the Lean implementation.

4 Explanatory Extension: Semantic Framework

4.1 Core Frame

Definition 4.1 (Core Frame)**.** A *core frame* is a structure $\mathbf{F} = \langle S, \sqsubseteq, D, \Rightarrow \rangle$ where:

- $\langle S, \sqsubseteq \rangle$ is a constitutive frame
- $D = \langle D, +, \leqslant \rangle$ is a totally ordered abelian group
- $\Rightarrow$ is a ternary relation on $S \times D \times S$ satisfying the constraints below

Notation 4.2. We write $s \Rightarrow_d t$ to denote that $(s, d, t) \in \Rightarrow$, read "there is a task from s to t of duration d."

4.2 State Modality Definitions

Definition 4.3 (State Modality)**.** Let $\mathbf{F} = \langle S, \sqsubseteq, D, \Rightarrow \rangle$ be a core frame.

$$s \in P \ \textit{iff}\ s \Rightarrow_0 s \qquad \text{(Possible state)} \tag{26}$$

$$s \notin P \ \textit{iff}\ \neg(s \Rightarrow_0 s) \qquad \text{(Impossible state)} \tag{27}$$

$$s \sim t \ \textit{iff}\ s \Rightarrow_d t \text{ or } t \Rightarrow_d s \text{ for some } d \qquad \text{(Connected states)} \tag{28}$$

$$s \circ t \ \textit{iff}\ s \sqcup t \in P \qquad \text{(Compatible states)} \tag{29}$$

$$w \in W \ \textit{iff}\ w \text{ is a maximal possible state} \qquad \text{(World-state)} \tag{30}$$

$$s \in N \ \textit{iff}\ s \sim t \text{ implies } s = t \qquad \text{(Necessary state)} \tag{31}$$

Definition 4.4 (Maximal Compatible Part)**.** The set of *maximal t-compatible parts* of r is:

$$r_t := \{s : s \sqsubseteq r, s \circ t, \text{ and } s' \sqsubseteq s \text{ for all } s' \text{ where } s \sqsubseteq s' \sqsubseteq r \text{ and } s' \circ t\}$$

Remark 4.5. A world-state $w \in W$ is maximal in the sense that $t \sqsubseteq w$ for every compatible state $t \circ w$.

4.3 Task Relation Constraints

Definition 4.6 (Task Relation Constraints)**.** The task relation $\Rightarrow$ satisfies:

1. **Compositionality**: If $s \Rightarrow_x t$ and $t \Rightarrow_y u$, then $s \Rightarrow_{x+y} u$
2. **Parthood (Left)**: If $d \sqsubseteq s$ and $s \Rightarrow_x t$, then $d \Rightarrow_x r$ for some $r \sqsubseteq t$
3. **Parthood (Right)**: If $r \sqsubseteq t$ and $s \Rightarrow_x t$, then $d \Rightarrow_x r$ for some $d \sqsubseteq s$
4. **Containment (Left)**: If $s \in P$, $d \sqsubseteq s$, and $d \Rightarrow_x r$, then $s \Rightarrow_x t \sqcup r$ for some $t \in S$
5. **Containment (Right)**: If $t \in P$, $r \sqsubseteq t$, and $d \Rightarrow_x r$, then $s \sqcup d \Rightarrow_x t$ for some $s \in S$
6. **Maximality**: If $s \in S$ and $t \in P$, there is a maximal t-compatible part $r \in s_t$

Remark 4.7. The Containment constraints ensure that tasks between parts of possible states can be extended to tasks between the states themselves. The Maximality constraint ensures that for any state and possible state, there exists a maximal part compatible with that possible state.

See `Logos.Explanatory.Frame.CoreFrame` for the Lean implementation.

4.4 World-History

Definition 4.8 (World-History)**.** A *world-history* over a core frame $\mathbf{F}$ is a function $\tau : X \to W$ where:

- $X \subseteq D$ is a convex subset of the temporal order
- $\tau(x) \Rightarrow_{y-x} \tau(y)$ for all times $x, y \in X$ where $x \leqslant y$

Notation 4.9. The set of all world-histories over $\mathbf{F}$ is denoted $H_{\mathbf{F}}$.

Remark 4.10. World-histories assign world-states to times in a way that respects the task relation. The constraint ensures that consecutive world-states are connected by appropriate tasks. The set of maximal possible evolutions equals the set of world-histories (proven in Brast-McKie, "Counterfactual Worlds").

4.5 Core Model

Definition 4.11 (Core Model)**.** A *core model* is a structure $\mathcal{M} = \langle S, \sqsubseteq, D, \Rightarrow, |\rangle|$ where:

- $\langle S, \sqsubseteq, D, \Rightarrow\rangle$ is a core frame
- $||$ is an interpretation as in the Constitutive Foundation

See `Logos.Explanatory.Model.CoreModel` for the Lean implementation.

4.6 Truth Conditions

Truth is evaluated relative to a model $\mathcal{M}$, world-history τ, time $x \in D$, variable assignment σ, and temporal index $\vec{\imath} = \langle i_1, i_2, \ldots\rangle$:

4.6.1 Atomic Sentences

Definition 4.12 (Atomic Truth)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash F(t_1, \ldots, t_n) \;\; \textit{iff} \;\; \exists i : \texttt{Fin}\ (\#^{+}(F)) : \mathrm{verfn}_F(i)(\lambda j : \texttt{Fin}\ n, [\![t_j]\!]^{\sigma}_{\mathcal{M}}) \sqsubseteq \tau(x) \tag{32}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \nvDash F(t_1, \ldots, t_n) \;\; \textit{iff} \;\; \exists i : \texttt{Fin}\ (\#^{-}(F)) : \mathrm{faln}_F(i)(\lambda j : \texttt{Fin}\ n, [\![t_j]\!]^{\sigma}_{\mathcal{M}}) \sqsubseteq \tau(x) \tag{33}$$

Remark 4.13. It is derivable that $\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash \varphi$ iff it is not the case that $\mathcal{M}, \tau, x, \sigma, \vec{\imath} \nvDash \varphi$. This justifies using $\vDash$ alone for truth and $\nvDash$ for falsehood.

4.6.2 Lambda Abstraction and Quantification

Definition 4.14 (Lambda and Quantification)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash (\lambda y.\varphi)(t) \;\; \textit{iff} \;\; \mathcal{M}, \tau, x, \sigma[[\![t]\!]^{\sigma}_{\mathcal{M}}/y], \vec{\imath} \vDash \varphi \tag{34}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash \forall y.\varphi \;\; \textit{iff} \;\; \mathcal{M}, \tau, x, \sigma[s/y], \vec{\imath} \vDash \varphi \text{ for all } s \in S \tag{35}$$

4.6.3 Actuality Predicate

Definition 4.15 (Actuality)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash \mathsf{Act}(t) \;\; \textit{iff} \;\; [\![t]\!]^{\sigma}_{\mathcal{M}} \sqsubseteq \tau(x)$$

Remark 4.16 (Restricted Quantification). The actuality predicate enables restricting quantification to actually existing objects:

$\forall y(\mathsf{Act}(y) \to \varphi)$ is true iff φ holds for all states that are parts of $\tau(x)$

The unrestricted universal quantifier validates the Barcan formulas ($\forall x \Box \varphi \to \Box \forall x \varphi$ and its converse), while the actuality-restricted quantifier does not.

4.6.4 Extensional Connectives

Definition 4.17 (Extensional Connectives)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \neg\varphi \ \textit{iff}\ \mathcal{M}, \tau, x, \sigma, \vec{\imath} \not\models \varphi \tag{36}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \wedge \psi \ \textit{iff}\ \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \text{ and } \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \psi \tag{37}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \vee \psi \ \textit{iff}\ \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \text{ or } \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \psi \tag{38}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \to \psi \ \textit{iff}\ \mathcal{M}, \tau, x, \sigma, \vec{\imath} \not\models \varphi \text{ or } \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \psi \tag{39}$$

4.6.5 Modal Operators

Definition 4.18 (Modal Operators)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \Box\varphi \ \textit{iff}\ \mathcal{M}, \alpha, x, \sigma, \vec{\imath} \models \varphi \text{ for all } \alpha \in H_{\mathbf{F}} \tag{40}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \Diamond\varphi \ \textit{iff}\ \mathcal{M}, \alpha, x, \sigma, \vec{\imath} \models \varphi \text{ for some } \alpha \in H_{\mathbf{F}} \tag{41}$$

Remark 4.19. $\Diamond\varphi \equiv \neg\Box\neg\varphi$. Metaphysical necessity can also be defined via counterfactuals: $\Box\varphi := \top \mathrel{\Box\!\!\rightarrow} \varphi$. This yields an S5 modal logic.

4.6.6 Core Tense Operators

Definition 4.20 (Core Tense Operators)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models H\varphi \ \textit{iff}\ \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for all } y \in D \text{ where } y < x \tag{42}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models G\varphi \ \textit{iff}\ \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for all } y \in D \text{ where } y > x \tag{43}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models P\varphi \ \textit{iff}\ \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for some } y \in D \text{ where } y < x \tag{44}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models F\varphi \ \textit{iff}\ \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for some } y \in D \text{ where } y > x \tag{45}$$

Remark 4.21 (Equivalences). $P\varphi \equiv \neg H\neg\varphi$ and $F\varphi \equiv \neg G\neg\varphi$.

4.6.7 Extended Tense Operators

Definition 4.22 (Since and Until)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \lhd \psi \ \textit{iff} \ \text{there exists } z < x \text{ where } \mathcal{M}, \tau, z, \sigma, \vec{\imath} \models \psi \text{ and } \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for all } y \text{ where } z < y < x \quad (46)$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \rhd \psi \ \textit{iff} \ \text{there exists } z > x \text{ where } \mathcal{M}, \tau, z, \sigma, \vec{\imath} \models \psi \text{ and } \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for all } y \text{ where } x < y < z \quad (47)$$

Remark 4.23. "φ since ψ" means ψ was true at some past time, and φ has been true ever since. "φ until ψ" means ψ will be true at some future time, and φ is true until then.

4.6.8 Counterfactual Conditional

Definition 4.24 (Counterfactual Conditional)**.**

$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \mathrel{\Box\!\!\rightarrow} \chi$ *iff* for all $i : \texttt{Fin}\ (\#^{+}(\varphi)), t := \text{verfn}_{\varphi}(i)$ and $\beta \in H_{\mathbf{F}}$:

if $s \sqcup t \sqsubseteq \beta(x)$ for some maximal t-compatible part $s \in \tau(x)_t$, then $\mathcal{M}, \beta, x, \sigma, \vec{\imath} \models \chi$

Definition 4.25 (Imposition)**.** We write $t \rightarrow_w w'$ ("imposing t on w yields w'") iff there exists a maximal t-compatible part $s \in w_t$ where $s \sqcup t \sqsubseteq w'$.

Remark 4.26 (Intuitive Reading). A counterfactual "if φ were the case, then χ" is true at world τ and time x iff the consequent χ is true in any world β at x where $\beta(x)$ is the result of minimally changing $\tau(x)$ to make the antecedent φ true.

4.6.9 Store and Recall Operators

Definition 4.27 (Store and Recall)**.** For cross-temporal reference within counterfactual evaluation, the context includes a temporal index $\vec{\imath} = \langle i_1, i_2, \ldots \rangle$ of stored times:

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \uparrow^{k} \varphi \ \textit{iff} \ \mathcal{M}, \tau, x, \sigma, \vec{\imath}[x/i_k] \models \varphi \quad (48)$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \downarrow^{k} \varphi \ \textit{iff} \ \mathcal{M}, \tau, i_k, \sigma, \vec{\imath} \models \varphi \quad (49)$$

where $\vec{\imath}[x/i_k]$ replaces i_k with current time x.

Example 4.28 (Tensed Counterfactuals)**.**

- $\downarrow^{1} (\psi \mathrel{\Box\!\!\rightarrow} FH)$ — "If ψ had occurred at the stored time, then H would have occurred at some future time"

- $\uparrow^2\downarrow^1\ (\psi \mathrel{\Box\!\!\rightarrow} \downarrow^2 H)$ — "Store now, recall past time, if ψ then H at the stored present"

See `Logos.Explanatory.Semantics.Satisfies` for the Lean implementation.

4.7 Bimodal Interaction Principles

The task semantics validates these perpetuity principles connecting modal and temporal operators:

Definition 4.29 (Perpetuity Principles)**.**

P1 :	$\Box\varphi \to \triangle\varphi$	Necessary implies always	(50)
P2 :	$\triangledown\varphi \to \Diamond\varphi$	Sometimes implies possible	(51)
P3 :	$\Box\triangle\varphi \leftrightarrow \triangle\Box\varphi$	Necessary-always commutativity	(52)
P4 :	$\Diamond\triangledown\varphi \leftrightarrow \triangledown\Diamond\varphi$	Possible-sometimes commutativity	(53)
P5 :	$\Box\varphi \to \Box\triangle\varphi$	Necessary implies necessary-always	(54)
P6 :	$\triangledown\Diamond\varphi \to \Diamond\varphi$	Sometimes-possible implies possible	(55)

4.8 Temporal Frame Constraints

Different temporal structures yield different valid principles. The framework does not assume discrete time:

Constraint	Description	Corresponding Axiom
Dense	Between any two times there is another	$GG\varphi \to G\varphi$
Complete	Every bounded set has a least upper bound	$\triangle(H\varphi \to FH\varphi) \to (H\varphi \to G\varphi)$
Unbounded Past	No earliest time	$P\top$
Unbounded Future	No latest time	$F\top$

Table 2: Temporal frame constraints and corresponding axioms

4.9 Logical Consequence

Definition 4.30 (Logical Consequence (Explanatory Extension))**.**

$\Gamma \vDash \varphi$ *iff* for any $\mathcal{M}, \tau \in H_{\mathbf{F}}, x \in D, \sigma, \vec{\imath}$: if $\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash \psi$ for all $\psi \in \Gamma$, then $\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash \varphi$

Remark 4.31. Unlike the Constitutive Foundation, the Explanatory Extension evaluates truth relative to world-histories and times, not just verification by states. The hyperintensional distinctions are preserved—propositions still differ if their exact verifiers differ—but the Explanatory Extension adds the intensional layer needed to evaluate sentences with modal and temporal operators.

5 Counterfactual Logic Axiom System

This section presents the axiom system for the counterfactual logic of the Explanatory Extension.

5.1 Core Rules

Definition 5.1 (Closure Under Deduction)**.**

$$\textbf{R1}: \quad \text{If } \Gamma \vdash \chi, \text{ then } \varphi \mathrel{\Box\!\!\rightarrow} \Gamma \vdash \varphi \mathrel{\Box\!\!\rightarrow} \chi$$

Remark 5.2. This rule states that if χ is derivable from Γ, then for any antecedent φ, the counterfactual with consequent χ is derivable from the counterfactuals with consequents from Γ.

5.2 Counterfactual Axiom Schemata

Definition 5.3 (Counterfactual Axioms)**.**

$$\textbf{C1}: \quad \varphi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Identity)} \tag{56}$$

$$\textbf{C2}: \quad \varphi, \varphi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \qquad \text{(Counterfactual Modus Ponens)} \tag{57}$$

$$\textbf{C3}: \quad \varphi \mathrel{\Box\!\!\rightarrow} \psi, \varphi \wedge \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Weakened Transitivity)} \tag{58}$$

$$\textbf{C4}: \quad \varphi \vee \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \wedge \psi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Disjunction-Conjunction)} \tag{59}$$

$$\textbf{C5}: \quad \varphi \vee \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Simplification Left)} \tag{60}$$

$$\textbf{C6}: \quad \varphi \vee \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \psi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Simplification Right)} \tag{61}$$

$$\textbf{C7}: \quad \varphi \mathrel{\Box\!\!\rightarrow} \varphi, \psi \mathrel{\Box\!\!\rightarrow} \varphi, \varphi \wedge \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \vee \psi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Disjunction Introduction)} \tag{62}$$

Remark 5.4 (Explanation of Counterfactual Axioms).

- **C1** (Identity): Every proposition counterfactually implies itself.
- **C2** (Modus Ponens): If φ is actually true and φ would imply φ, then φ is true.
- **C3** (Weakened Transitivity): This weaker form of transitivity avoids the problematic full transitivity principle.
- **C4-C7**: These axioms govern the interaction between counterfactuals and Boolean operations on antecedents.

5.3 Modal Axiom Schemata

Definition 5.5 (Modal Axioms)**.**

M1 :	$\top$	(Truth)	(63)
M2 :	$\neg\bot$	(Non-Contradiction)	(64)
M3 :	$\varphi \to \Box\Diamond\varphi$	(Brouwer)	(65)
M4 :	$\Box\varphi \to \Box\Box\varphi$	(S4 Transitivity)	(66)
M5 :	$\Box(\varphi \to \varphi) \vdash \varphi \mathrel{\Box\!\!\to} \varphi$	(Strict-to-Counterfactual)	(67)

Remark 5.6 (S5 Modal Logic). The combination of M3 and M4 yields an S5 modal logic, which is characterized by an equivalence relation on possible worlds. In the task semantics, this arises from the structure of world-histories: any world-history is accessible from any other at any given time.

Remark 5.7 (Strict-to-Counterfactual). Axiom M5 states that if φ is a strict consequence of φ (i.e., necessarily, if φ then φ), then φ counterfactually implies φ. This connects the modal operator with the counterfactual conditional.

5.4 Derived Theorems

Theorem 5.8 (Counterfactual Strengthening)**.** *If $\varphi \mathrel{\Box\!\!\to} \varphi$ and $\varphi \mathrel{\Box\!\!\to} \psi$, then $\varphi \mathrel{\Box\!\!\to} (\varphi \wedge \psi)$.*

Theorem 5.9 (Counterfactual Weakening)**.** *If $\varphi \mathrel{\Box\!\!\to} \varphi$ and $\varphi \vdash \psi$, then $\varphi \mathrel{\Box\!\!\to} \psi$.*

Theorem 5.10 (Necessity Introduction)**.** $\top \mathrel{\Box\!\!\to} \varphi$ *is equivalent to* $\Box\varphi$.

Theorem 5.11 (Excluded Middle)**.** $(\varphi \mathrel{\Box\!\!\to} \varphi) \vee (\varphi \mathrel{\Box\!\!\to} \neg\varphi)$ *is* not *generally valid.*

Remark 5.12. The failure of excluded middle for counterfactuals (Theorem 5.11) reflects the mereological structure: when imposing an antecedent φ on the actual world, there may be multiple ways to do so, leading to different worlds where φ or $\neg\varphi$ holds. The counterfactual is true only when the consequent holds in *all* such imposed worlds.

5.5 Soundness and Completeness

Theorem 5.13 (Soundness)**.** *If $\Gamma \vdash \varphi$ using the axiom system above, then $\Gamma \models \varphi$.*

> [OPEN QUESTION]: Completeness of the axiom system with respect to the mereological counterfactual semantics remains an open problem. The main challenge is showing that every consistent set of formulas has a model with the required mereological structure.

See `Logos.Explanatory.Axioms.CounterfactualAxioms` for the Lean implementation.

6 Epistemic Extension

[DETAILS PENDING DEVELOPMENT]

The Epistemic Extension extends the Explanatory Extension with structures for belief, knowledge, and probability.

6.1 Frame Extension

[DETAILS PENDING DEVELOPMENT]

The epistemic frame extends the core frame with a credence function assigning probabilities to state transitions.

> [OPEN QUESTION]: What is the exact structure of the credence function? Does it assign probabilities to individual state transitions or to sets of transitions?

6.2 Operators

[FULL SEMANTIC CLAUSES FOR EPISTEMIC OPERATORS PENDING SPECIFICATION]

Operator	Intended Reading
$B_a\varphi$	Agent a believes that φ
$K_a\varphi$	Agent a knows that φ
$\Pr(\varphi) \geqslant r$	The probability of φ is at least r

Table 3: Epistemic operators

6.3 Indicative Conditionals

[DETAILS PENDING DEVELOPMENT]

> [OPEN QUESTION]: How do indicative conditionals relate to counterfactual conditionals in the semantic framework?

7 Normative Extension

[DETAILS PENDING DEVELOPMENT]

The Normative Extension extends the Explanatory Extension with structures for obligation, permission, and value.

7.1 Frame Extension

[DETAILS PENDING DEVELOPMENT]

The normative frame extends the core frame with value orderings over states.

> [OPEN QUESTION]: How are value orderings structured? Are they complete orderings or partial orderings? Are they agent-relative?

7.2 Operators

Operator	Intended Reading
$O\varphi$	It ought to be that φ
$P\varphi$	It is permitted that φ
$\varphi \succ \psi$	φ is preferred to ψ

Table 4: Normative operators

[FULL SEMANTIC CLAUSES FOR NORMATIVE OPERATORS PENDING SPECIFICATION]

7.3 Normative Explanation

[DETAILS PENDING DEVELOPMENT]

> [OPEN QUESTION]: How does normative explanation relate to causal explanation?

8 Spatial Extension

[DETAILS PENDING DEVELOPMENT]

The Spatial Extension extends the Explanatory Extension with structures for spatial reasoning and location.

8.1 Frame Extension

[DETAILS PENDING DEVELOPMENT]

The spatial frame extends the core frame with:

- **Location space** L = set of spatial locations
- **Spatial relations**: adjacency, containment, distance

> [OPEN QUESTION]: What spatial primitives are required? Should locations be mereological (with parts) or set-theoretic?

8.2 Operators

Operator	Intended Reading
Here(φ)	φ holds at the current location
Somewhere(φ)	φ holds at some location
Everywhere(φ)	φ holds at all locations
Near(φ)	φ holds at an adjacent location

Table 5: Spatial operators

[FULL SEMANTIC CLAUSES FOR SPATIAL OPERATORS PENDING SPECIFICATION]

9 Agential Extension

[DETAILS PENDING DEVELOPMENT]

The Agential Extension requires at least one of the Epistemic, Normative, or Spatial Extensions to be loaded. It provides structures for multi-agent reasoning.

Remark 9.1 (Dependency). This extension depends on at least one middle extension (Epistemic, Normative, or Spatial) being loaded. See the extension dependency diagram in section 1.1.

9.1 Frame Extension

[DETAILS PENDING DEVELOPMENT]

> [OPEN QUESTION]: What frame extensions are required for multi-agent reasoning? Does this extension add agent indices, or agent-relative accessibility relations?

9.2 Multi-Agent Operators

[DETAILS PENDING DEVELOPMENT]

> [OPEN QUESTION]: How do individual and collective agency interact in the semantic framework?

10 Reflection Extension

The Reflection Extension enables first-person metacognitive reasoning through the I operator. Where the Agential Extension projects epistemic operators onto other agents, the Reflection Extension projects them onto the self, enabling perspectival distance from one's own beliefs, abilities, and goal-states.

Remark 10.1 (Dependency). This extension inherits from the Epistemic Extension in parallel with the Agential Extension. Both apply the same epistemic apparatus: Agential projects outward onto other agents, Reflection projects inward onto self. See the extension dependency diagram in section 1.1.

10.1 Core Insight

The I operator transforms direct modal expressions into self-aware epistemic stances. Consider the difference between:

Expression	Reading	Epistemic Subject
$\mathsf{Mu}(\varphi)$	"It must be that φ"	Implicit/absent
$\mathsf{I}(\varphi)$	"I believe that φ"	Explicit (self)

The direct expression commits the speaker without self-awareness; the self-aware expression registers the speaker as the epistemic subject. This distinction follows Kaplan's character/content framework and Lewis's centered-worlds semantics.

10.2 Frame Extension

The Reflection frame extends the Agential frame with:

Component	Description
Self-Index	Distinguished agent index $\mathsf{self} \in A$
Self-Accessibility	Reflexive accessibility relation R_{self}
Self-Model	Function $\mathsf{SM} : W \to \mathsf{SelfState}$
Commitment Register	Set $\mathsf{CR}(w)$ of self-ascribed propositions at w

The self-accessibility relation R_{self} satisfies:

- **Seriality (D)**: For all w, there exists w' such that $(w, w') \in R_{\mathsf{self}}$
- **Transitivity (4)**: Positive introspection: $\mathsf{I}(\varphi) \to \mathsf{I}(\mathsf{I}(\varphi))$
- **Euclidean (5)**: Negative introspection: $\neg\mathsf{I}(\varphi) \to \mathsf{I}(\neg\mathsf{I}(\varphi))$

10.3 Truth Conditions

Given evaluation context $(\mathcal{M}, \tau, x, \sigma, \vec{i})$, the primary I operator is defined by:

Definition 10.2 (Truth Condition for I)**.** $\mathcal{M}, \tau, x, \sigma \models \mathsf{I}(\varphi)$ if and only if:

1. For all w' accessible via R_{self} from $\tau(x)$: $\mathcal{M}, \tau[w'/x], x, \sigma \models \varphi$
2. $\varphi \in \mathsf{CR}(\tau(x))$

Remark 10.3 (Commitment Register Distinction). The second condition distinguishes explicit self-attribution from mere truth across epistemic alternatives. A proposition may hold at all worlds compatible with the agent's

epistemic state without the agent having explicitly endorsed it. The Commitment Register tracks which propositions the agent has actively self-ascribed, capturing the difference between implicit epistemic commitment and explicit metacognitive awareness.

10.4 Operators

10.4.1 Self-Knowledge and Self-Belief

Operator	Symbol	Reading
Metacognitive I	$\mathsf{I}(\varphi)$	"I judge/believe that φ"
Self-Knowledge	$\mathsf{I}_K(\varphi)$	"I know that φ"
Self-Belief	$\mathsf{I}_B(\varphi)$	"I believe that φ"
Self-Uncertainty	$\mathsf{I}_?(\varphi)$	"I am uncertain whether φ"

10.4.2 Ability Introspection

Operator	Symbol	Reading
Self-Ability	$\mathsf{I}_{\mathsf{Can}}(\varphi)$	"I can bring about φ"
Ability-Knowledge	$K_{\mathsf{self}}(\mathsf{Can}_{\mathsf{self}}(\varphi))$	"I know I can do φ"
Limitation-Knowledge	$K_{\mathsf{self}}(\neg\mathsf{Can}_{\mathsf{self}}(\varphi))$	"I know I cannot do φ"

10.4.3 Goal-Distance

Operator	Symbol	Reading
Goal-Distance	$\mathsf{Dist}(G, n)$	"Goal G is n steps away"
Goal-Progress	$\mathsf{Closer}(G)$	"I am getting closer to G"
Goal-Achievable	$\mathsf{Achievable}(G)$	"Goal G is achievable"

10.4.4 Attributed Belief

Operator	Symbol	Reading
Other's Belief About Self	$B_j(\mathsf{I}(\varphi))$	"Agent j believes I believe φ"
Other's Belief About Ability	$B_j(\mathsf{I}_{\mathsf{Can}}(\varphi))$	"Agent j believes I can do φ"

10.4.5 Derived Operators

The following derived operators are defined in terms of the primary I operator:

Operator	Definition	Reading
$\mathsf{I}_K(\varphi)$	$\mathsf{I}(K(\varphi))$	"I know that I know φ"
$\mathsf{I}_B(\varphi)$	$\mathsf{I}(B(\varphi))$	"I know that I believe φ"
$\mathsf{I}_?(\varphi)$	$\mathsf{I}(\neg B(\varphi) \wedge \neg B(\neg\varphi))$	"I know I am uncertain about φ"
$\mathsf{I}_{\mathsf{Can}}(\varphi)$	$\exists \vec{a}\, \mathsf{Achievable}_{\mathsf{self}}(\vec{a}) \wedge \mathsf{LeadsTo}(\vec{a}, \varphi)$	"I can bring about φ"

Remark 10.4 (Self-Knowledge vs. Self-Belief). The operators I_K and I_B differ in their factivity requirements. $\mathsf{I}_K(\varphi)$ requires that the inner knowledge claim $K(\varphi)$ be factive (hence φ is true), while $\mathsf{I}_B(\varphi)$ requires only that the agent believes φ. Self-uncertainty $\mathsf{I}_?(\varphi)$ captures metacognitive awareness of one's own agnosticism.

10.5 Key Axioms

Axiom	Name	Schema
I-T	Veridicality	$\mathsf{I}_K(\varphi) \to \varphi$
I-4	Positive Introspection	$\mathsf{I}(\varphi) \to \mathsf{I}(\mathsf{I}(\varphi))$
I-5	Negative Introspection	$\neg\mathsf{I}(\varphi) \to \mathsf{I}(\neg\mathsf{I}(\varphi))$
I-D	Consistency	$\neg\mathsf{I}(\bot)$
I-Commit	Commitment Closure	$\mathsf{I}(\varphi) \to \mathsf{Register}(\varphi)$

10.6 Temporal Interaction

The metacognitive I operator interacts non-trivially with temporal operators. The following formulas are semantically distinct:

Formula	Reading
$P(\mathsf{I}(\varphi))$	"I previously believed φ"
$\mathsf{I}(P(\varphi))$	"I believe φ was the case"
$F(\mathsf{I}(\varphi))$	"I will believe φ"
$\mathsf{I}(F(\varphi))$	"I believe φ will be the case"

Remark 10.5 (Non-Equivalence). The pairs $P(\mathsf{I}(\varphi))$ and $\mathsf{I}(P(\varphi))$ are not equivalent. The former describes a past epistemic state (what was believed then), while the latter describes a current belief about the past (what is believed now about then). An agent may currently believe something was true without having believed it at the time, and may have believed something without currently believing it was true. This demonstrates the interaction between temporal and metacognitive perspectives.

10.7 Relationship to Agential Extension

The Reflection Extension inherits from the Epistemic Extension in parallel with the Agential Extension:

$$\begin{tikzcd}[column sep=small] & \text{Epistemic Extension} \arrow[dl, "\text{outward}"'] \arrow[dr, "\text{inward}"] & \\ \text{Agential} & & \text{Reflection} \end{tikzcd}$$

Both extensions apply the epistemic apparatus in different directions:

- **Agential**: $B_j(\varphi)$ – "Agent j believes φ" (modeling others)
- **Reflection**: $\mathsf{I}(\varphi)$ – "I believe φ" (modeling self)

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